EXACT ENUMERATION / DATA / VERIFICATION

Rank-4 fusion rings

A reproducible census through multiplicity 128

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Abstract

We explain the exact enumeration of all based fusion rings of rank 4 through multiplicity 128, comprising 889,530 isomorphism classes. The account separates the mathematical coverage argument, the completed search, and independent verification of the emitted tensors. All duality types are included, and no categorifiability filter is applied. Code, machine-readable counts, full multiplication tensors and provenance records accompany the text.

AI-generated research · independent verification pending. This manuscript, its code and data were developed with AI assistance. Independent human verification and Lean formalization are pending. Treat the results with caution: exact computational checks establish only their stated scope.

Scope. A census of based rings, not a classification of tensor categories. Only completed whole-rank results are released.

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1 Objects, equivalence and the counting convention

A *fusion ring* here is a free abelian group with a distinguished finite basis $B = \{b_0, b_1, \dots, b_{r-1}\}$, an associative multiplication with unit b_0, and an involution $i \mapsto i^*$ fixing 0. Its structure constants satisfy

$$b_i b_j = \sum_k N_{ij}^k b_k, \qquad N_{ij}^k \in \mathbb{Z}_{\geq 0},$$

$$N_{0j}^k = N_{j0}^k = \delta_{jk}, \qquad N_{ij}^0 = \delta_{i^*,j}, \tag{1}$$

$$N_{ij}^k = N_{i^*k}^j = N_{kj^*}^i. \tag{2}$$

These conventions agree with the usual based-ring definition; see [1, 2]. No assumption of categorifiability, commutativity or integrality of Frobenius–Perron dimensions is imposed unless proved in the relevant rank.

A based isomorphism is a bijection of the distinguished bases preserving the unit and multiplication. In tensor notation it is a permutation q fixing 0 such that

$$N_{ij}^k = N'^{q(k)}_{q(i)q(j)} \quad \text{for every } i, j, k.$$

The coefficient of the unit determines the involution, so such an isomorphism automatically intertwines duality. The *multiplicity* is the maximum entry of the *full* ordered multiplication tensor. It is at least one, including for group rings. Put

$$c_r(m) = \#\{\text{rank-}r \text{ fusion rings of multiplicity exactly } m\}/\cong, \qquad C_r(M) = \sum_{m=1}^{M} c_r(m).$$

A list through M contains all rings of multiplicity at most M, not only those of multiplicity equal to M. This distinction is essential when exporting OEIS b-files.

1.1 Why duality types can be searched separately

An involution on the $r-1$ nonunit elements consists of p disjoint transpositions and $r-1-2p$ fixed points, with $0 \leq p \leq \lfloor (r-1)/2 \rfloor$. Fix one representative d_p of each type. An isomorphism between two tensors with involution d_p commutes with d_p. Hence it is sufficient to quotient that stratum by the centralizer

$$Z_{S_{r-1}}(d_p), \qquad |Z_{S_{r-1}}(d_p)| = 2^p p!(r-1-2p)!.$$

Different p cannot be isomorphic. Therefore a whole-rank census is complete exactly when every required stratum has been searched exhaustively. Finishing only the non-self-dual strata does not certify a whole-rank counting term.

2 The certified release at rank 4

The released bound is $M = 128$ and the cumulative number of classes is $C_4(128) = 889,530$. Every duality type is included. The accompanying full tensor file has one record per based-isomorphism class.

Exact multiplicity m	$c_4(m)$
1	10
2	17
3	24
4	45
5	55
6	81
7	92
8	137
9	151
10	186
11	238
12	291
13	246
14	340
15	349
16	525

The complete table through the released bound appears in Appendix A.

The data and their checksums are recorded in `results/census.json`. The corresponding completed-run evidence and independent audit are under `verification/rank4/`.

3 Why the rank-four specialization is commutative

The coefficient-of-unit functional τ gives the positive inner product $\langle x, y\rangle = \tau(xy^*)$ on the complexified based ring. The basis is orthonormal and multiplication is closed under adjoints. Consequently its complexification is semisimple. The Frobenius–Perron dimension character is a one-dimensional representation. A noncommutative semisimple complex algebra with a one-dimensional representation has dimension at least $1 + 2^2 = 5$. Thus a rank-four fusion ring is commutative. This justifies the specialization below without any categorical assumption.

4 An independent rank-four census

We give a complete integer enumeration, with no input from an external catalogue. In this section all parameters lie in $\{0, \dots, M\}$. The unit coefficients ensure that the multiplicity is the maximum of one and the parameters. Commutativity was proved above; duality has just two possible types.

4.1 Self-dual parameters and elimination

Frobenius reciprocity gives the following general matrices:

$$N_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & c \\ 0 & b & d & e \\ 0 & c & e & f \end{pmatrix}, \quad N_y = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & b & d & e \\ 1 & d & g & h \\ 0 & e & h & i \end{pmatrix}, \quad N_z = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & c & e & f \\ 0 & e & h & i \\ 1 & f & i & j \end{pmatrix}.$$

Their pairwise commutativity is equivalent to the six equations

$$ad - b^2 + bg + ch - d^2 - e^2 + 1 = 0, \tag{3}$$
$$ae - bc + bh + ci - de - ef = 0, \tag{4}$$
$$be - cd + dh - eg + ei - fh = 0, \tag{5}$$
$$af + bi - c^2 + cj - e^2 - f^2 + 1 = 0, \tag{6}$$
$$bf - ce + di - eh + ej - fi = 0, \tag{7}$$
$$df - e^2 + gi - h^2 + hj - i^2 + 1 = 0. \tag{8}$$

These are necessary and sufficient for associativity. Indeed the first row of N_uN_v records uv. Commutativity implies that every other row of $N_uN_v - \sum_w N_{uv}^w N_w$ is obtained by multiplying its zero first row by a regular matrix, so the difference vanishes.

The six mixed coordinates b, c, d, f, h, i form one orbit under permutations of x, y, z. Every based-ring orbit therefore has a representative with

$$b = \min\{b, c, d, f, h, i\}.$$

For $b > 0$, enumerate b, c, d, e, f, with $c, d, f \geq b$, and put

$$H_0 = e(d+f) + bc, \qquad G_0 = b^2 + d^2 + e^2 - 1,$$
$$J_0 = c^2 + e^2 + f^2 - 1, \qquad A = e(b^2 - c^2) + bc(f - d),$$
$$C = e(bf - ce), \qquad B_0 = b^3e - b^2cd + b(d-f)H_0 - ebG_0 + ecH_0.$$

The first, second and fourth equations give

$$h = \frac{H_0 - ae - ci}{b}, \qquad g = \frac{G_0 - ad - ch}{b}, \qquad j = \frac{J_0 - af - bi}{c}.$$

After substitution, the third equation multiplied by b^2 is

$$Ca + Ai + B_0 = 0. \tag{9}$$

If $A \neq 0$, let $q = \gcd(A, C)$. There is no solution unless $q \mid B_0$. Otherwise a runs through the unique residue class solving $(C/q)a \equiv -B_0/q \pmod{|A/q|}$, and i is determined. A modulus of one means every bounded a. If $A = 0 \neq C$, determine a and enumerate i. If $A = C = 0$, require $B_0 = 0$ and retain all bounded a, i. Impose $0 \leq a, g, j \leq M$ and $b \leq h, i \leq M$, exact divisibility, and all six original equations.

If $b = 0 < c$, determine

$$h = \frac{d^2 + e^2 - ad - 1}{c}, \quad i = \frac{e(d + f - a)}{c}, \quad j = \frac{c^2 + e^2 + f^2 - af - 1}{c}.$$

For $e > 0$, integrality of i gives $a \equiv d + f \pmod{c/\gcd(c, e)}$. Enumerate c, d, e, f and this progression for a; the bounds on h, i, j give further linear interval restrictions. The equation $eg = h(d - f) + ei - cd$ determines g. For $e = 0$, nonnegativity forces $d, h > 0$ and

$$\gcd(c, d) = 1, \qquad a \equiv d - d^{-1} \pmod{c}, \qquad f = d - \frac{cd}{h}.$$

Indeed $ch = d^2 - ad - 1$ implies the coprimality, and $h(d - f) = cd$ gives the last identity. Enumerate c, d, a, determine h, f, j, and retain every bounded g. The congruence modulo one is unrestricted.

Finally, $b = c = 0$ gives

$$ad - d^2 - e^2 + 1 = 0, \quad e(a - d - f) = 0, \quad af - e^2 - f^2 + 1 = 0.$$

For $e > 0$ these are $a = d + f$ and $df = e^2 - 1$. Enumerate the bounded factor pairs, including $e = 1, d = 0$ with every bounded f, then enumerate h, i and determine

$$eg = h(d - f) + ei, \qquad ej = eh + i(f - d).$$

For $e = 0$, the preliminary equations force $a = 0, d = f = 1$. The remaining equation is $gi + hj = h^2 + i^2 - 2$. If $h > 0$, enumerate h, i, g and solve for j; if $h = 0 < i$, solve for g and retain every bounded j. The case $h = i = 0$ is impossible. In every branch, all six equations and all coefficient bounds are tested again.

4.2 One dual pair

If $X^* = X$ and $Y^* = Z$, the general matrices are

$$N_X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & b \\ 0 & b & c & d \\ 0 & b & d & c \end{pmatrix}, \quad N_Y = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & b & c & d \\ 0 & d & e & f \\ 1 & c & e & e \end{pmatrix}, \quad N_Z = N_Y^{\mathsf{T}}.$$

Associativity is equivalent to

$$\begin{aligned} ac - b^2 + 2be - c^2 - d^2 + 1 &= 0, & ad - b^2 + be + bf - 2cd &= 0, \\ b(c - d) + d(e - f) &= 0, & -c^2 + d^2 - e^2 + f^2 - 1 &= 0. \end{aligned}$$

Enumerate c, d, e. The last equation determines the unique nonnegative candidate $f = \sqrt{1 + c^2 - d^2 + e^2}$; retain it only if integral and bounded. If $c \neq d$, determine $b = -d(e - f)/(c - d)$. If $c = d$, require $d(e - f) = 0$ and retain every bounded b. Determine a from the first equation if $c > 0$, or from the second if $d > 0$; if $c = d = 0$, retain all bounded a. Finally test all four equations and all bounds. The square-root implementation corrects its initial estimate by integer comparisons and checks its square exactly.

Proposition 4.1 (Completeness and cost)**.** *The preceding algorithm enumerates all rank-four fusion rings of multiplicity at most M, up to based-ring isomorphism. The parameter search uses $O((M + 1)^6)$ integer operations before canonicalization.*

Proof. The two presentations exhaust duality and reciprocity. Each branch either solves a necessary linear equation by exact division or exhausts every allowed value when its coefficient vanishes. Final equation checks are necessary and sufficient for associativity. Thus no ring is omitted or spuriously admitted. In the self-dual search choose the lexicographically least parameter tuple among its permutations satisfying the same minimal-b condition. Comparing only those images is essential: the unconstrained least tuple need not satisfy that condition. Each orbit has exactly one retained tuple. In the other duality type the only possible nontrivial based automorphism interchanges Y, Z, and fixes all six parameters. Thus every retained non-self-dual tuple is already one class. The full-tensor interface additionally canonicalizes under all six permutations and checks distinctness independently.

There are five outer parameters in the positive-b branch. Solving (9) takes $O(\log(M+1))$ arithmetic operations for the greatest common divisor and at most $M + 1$ candidates. The apparently singular branch $A = C = B_0 = 0$ cannot occur there: if $e = 0$, then $A = 0$ forces $f = d$ and $B_0 = -b^2cd \neq 0$; if $e > 0$, the equations $A = C = 0$ give $cd = be$, $bf = ce$, and substitution gives $B_0 = be \neq 0$. The zero-b branches use at most five free coordinates, and the non-self-dual search at most five. The stated $O((M + 1)^6)$ arithmetic bound follows. Bit costs grow with $\log(M + 1)$; the implementation restricts its integer range explicitly. □

4.3 Current implementation and complete run

The current implementation is `code/rank4_census.cpp`; the reproduction entry point is `code/run_census.py`. The full run through multiplicity 128 produces 889530 based-isomorphism classes. Every output tensor is independently checked for the full fusion-ring axioms and canonicalized over all six unit-fixing permutations. The exact counts and execution reports are in the accompanying release. Exhaustiveness rests on the argument above, not on comparison with an existing list.

5 What the independent verification establishes

The enumerator and verifier have separate mathematical tasks. Exhaustiveness uses the complete coordinate parameterization, the safe-pruning proof and the normal completion of every required branch. The verifier reads only the emitted *full tensors*; it neither reconstructs the parameterization nor trusts the generator's polynomial list.

For every tensor it checks nonnegative integral coefficients, both unit laws, uniqueness and involutivity of the dual, both reciprocity identities, and every integer associativity identity for all four indices. It then canonicalizes under *every* unit-fixing basis permutation. A matching hash is only a lookup aid: actual canonical tensors are compared. This detects both literal repetitions and different labellings of an isomorphic based ring.

Passing these checks proves soundness and uniqueness of the emitted data; by itself it would not prove that no ring was omitted. That is why normal completion logs and the mathematical coverage argument are retained separately. Comparison with the original ancillary file is an additional containment/regression test, not an input to enumeration and not a replacement for an exhaustiveness proof.

5.1 Provenance and reproducibility

The retained Vercleyen–Slingerland v4 ancillary file has SHA-256

`b587c3f683157369da7cc090f762bbff0e18dcf11acc3f5031d6bc0c643843f8`

Its 28,451 records contain 25,138 distinct classes. The 3,313 redundant records occur at rank five and exact multiplicities 9–12. Their original and corrected counts are respectively

$$(1463, 1794, 2283, 3049) \quad \longrightarrow \quad (863, 1082, 1383, 1948).$$

Every deletion has an explicit basis-permutation witness in the audit folder. This audit does not establish completeness of the source's partial-search cells.

The primary manifest `results/census.json` records every released count, bound, filename and checksum. Compressed tensors use the original line-per-ring, nested-curly-brace format. An entry is $N[i][j][k] = N_{ij}^{k}$: the output index has already been dualized where the internal parameter convention requires it. The raw tensor needs no additional index conversion.

5.2 Certified datasets and reproducibility

The primary manifest gives the complete bound, exact counts and data checksums for every rank. Each `verification/rankR/` directory records the independent audit and source identity for that dataset. For the laptop censuses it also retains the completed enumeration, commands, environment, phase timings and nonempty subprocess logs. Every required duality stratum completed normally; independent canonicalization found zero duplicate classes.

The independent checks of ranks five through eight ran on GitHub using the supplied tensors. They did not rerun enumeration. The per-rank reports identify the exact verification revision and run, and certify agreement with the established exact-multiplicity count prefix. See `verification/README.md` for the evidence index and `docs/PROVENANCE.md` for attribution.

The local launcher has no clock limit, while the manual hosted workflow uses a finite shared deadline covering every computational phase. A timeout never certifies the unfinished bound. Only a complete, independently verified whole-rank candidate can enter the release; a later failed attempt cannot replace it. Reproduction commands and supported limits are in `docs/REPRODUCIBILITY.md`.

6 Reading and reusing the data

The result files count based rings, not monoidal categories or realizations of a ring. One tensor can admit several inequivalent categorifications, or none. No number here is a classification of fusion categories. The rank-specific count files use exact multiplicity; cumulative totals belong only in the summary column. Proposed OEIS additions must be regenerated from a complete manifest, not copied from an interrupted log.

The retained code was generated with AI assistance and is accompanied by explicit mathematical coverage arguments and independent integer checks. These checks are not a formal proof in a proof assistant. The present text is an explanatory computational companion, not a claim of journal acceptance or external peer review.

7 Reproduction commands

From the repository root on a local computer, request a complete run with no clock limit. An interruption does not certify a smaller census.

```
python3 scripts/run_laptop.py --rank 4 --bound 128
```

To verify the supplied release without recomputing it:

```
python3 scripts/check_release.py --full
```

For a shared-budget frontier search, use `scripts/extend_census.py`. For ranks five through eight it starts at the released bound plus one and restarts the unfinished bound without claiming checkpoint resumption. The local command above verifies all tensors and the old count prefix before packaging a result. The separate timed frontier wrapper bounds every phase. Rank three is instead capped at 1000.

A All exact-multiplicity rank-four counts

m	$c_4(m)$	m	$c_4(m)$	m	$c_4(m)$	m	$c_4(m)$
1	10	2	17	3	24	4	45
5	55	6	81	7	92	8	137
9	151	10	186	11	238	12	291
13	246	14	340	15	349	16	525
17	424	18	477	19	513	20	713
21	704	22	668	23	654	24	1,121
25	868	26	938	27	989	28	1,286
29	1,356	30	1,347	31	1,220	32	1,671
33	1,460	34	1,579	35	1,613	36	2,301
37	1,646	38	1,824	39	2,111	40	2,653
41	2,131	42	2,588	43	2,169	44	2,857
45	2,699	46	2,639	47	2,452	48	3,985
49	2,947	50	3,314	51	3,378	52	3,776
53	3,250	54	3,704	55	3,785	56	5,213
57	4,182	58	3,863	59	3,987	60	6,004
61	4,383	62	4,472	63	5,150	64	6,557
65	5,107	66	5,773	67	4,791	68	6,260
69	6,001	70	6,205	71	5,799	72	8,710
73	5,858	74	6,223	75	7,280	76	9,507
77	6,953	78	7,917	79	6,933	80	10,208
81	8,031	82	7,939	83	7,408	84	10,993
85	8,464	86	8,568	87	9,088	88	11,767
89	9,145	90	10,765	91	9,896	92	11,188
93	10,385	94	10,153	95	10,048	96	15,447
97	10,003	98	11,249	99	12,082	100	14,037
101	11,361	102	13,068	103	10,861	104	16,150
105	14,958	106	12,338	107	12,102	108	16,509
109	12,849	110	14,390	111	15,119	112	18,964
113	13,293	114	16,769	115	14,900	116	17,484
117	16,252	118	14,963	119	16,058	120	25,126
121	16,103	122	15,670	123	17,292	124	19,682
125	17,760	126	20,720	127	17,015	128	23,185

References

[1] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society, 2015.

[2] G. Vercleyen and J. Slingerland, *On Low Rank Fusion Rings*, Journal of Mathematical Physics 64 (2023), 091703; arXiv:2205.15637v4.

[3] V. Ostrik, *On formal codegrees of fusion categories*, Mathematical Research Letters 16 (2009), 895–901; arXiv:0810.3242.

[4] J. Dong and S. Palcoux, rank-four and rank-five census derivations, project manuscripts and computational supplements, September 2026. The relevant derivations are retained in the repository's `methods/` directory.

[5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entries A348305, A354471, A354472, A354473, A354475, A354476 and A354477. Proposed updates in this repository are not submitted edits.